

Nikhil Chaudhari

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Computer vision and interpretable AI researcher with a strong background in developing novel model architectures that improve explainability. My work spans end-to-end pipeline development for complex visual data, encompassing automated segmentation and 3D reconstruction. I bring deep experience in experimental design, performance optimization, and memory-efficient inference, enabling scalable deployment of state-of-the-art models. I collaborate effectively with cross-functional engineering teams to bridge research innovations with practical platform features and technical publications.

EDUCATION

University of Southern California <i>Ph.D., Biomedical Engineering</i>	Los Angeles, California, USA, 2021 - 2026
University of Southern California <i>M.S., Computer Science</i>	Los Angeles, California, USA, 2017 - 2019
Vellore Institute of Technology <i>B.Tech., Computer Science and Engineering</i>	Vellore, Tamil Nadu, India, 2013 - 2017

SKILLS

Machine Learning & Deep Learning: PyTorch, TensorFlow, diffusion models, ControlNet, generative modeling, DNN architecture design, regression, classification, large-scale model training on NVIDIA A100 GPUs

Research & Experimentation: Independent research agenda development, problem formulation, hypothesis testing, experimental design, statistical analysis, first-author publication track record

ML Systems & Optimization: CUDA/GPU programming, distributed training, performance optimization, memory-efficient inference, end-to-end ML pipeline development, GPU cluster administration, cost-aware resource optimization

Model Deployment & Inference: ONNX export/validation, ONNX Runtime (onnxruntime), TensorFlow/PyTorch-to-ONNX conversion, GPU-accelerated inference workloads, REST/API integration, client-server preprocessing

Interpretability & Reproducibility: Explainable AI (xAI), integrated gradients, SHAP, explainable model design, reproducible experimentation, open-source tool development [3]

Programming & Software Engineering: Python, C++, Java, MATLAB, Linux/bash, OOP, Git, GitHub Actions, CI/CD, production-ready code development

Cloud, DevOps & Infrastructure: AWS (EC2, S3), Google Cloud (GCP), Docker, Docker Compose orchestration, CI/CD-ready pipelines, GPU-accelerated container workloads

Data Engineering & Analytics: pandas, NumPy, ETL pipelines, SQL, Elasticsearch, Kibana, JSON, Oracle 11g

ML Applications & Visualization: Gradio, Streamlit, Plotly, HTML5 Canvas 2D rendering, JavaScript/ES modules, interactive ML application development

Collaboration & Technical Communication: Cross-functional research collaboration, project ownership and autonomy, translating complex research into practical solutions

SELECTED PROJECTS

- *Stanford Cars Classification & Model Explainability.* Developed an end-to-end PyTorch pipeline for a ResNet50 classifier on a 196-class dataset. Implemented and compared multiple explainable AI (xAI) techniques, including Grad-CAM, Integrated Gradients, LIME, and SHAP, to evaluate feature attribution. Built an interactive Streamlit application for real-time inference and dynamic saliency map visualization. [GitHub Demo](#)
- *Automated Video Analysis Pipeline.* Engineered an automated computer vision workflow to extract significant events from continuous video footage using pixel differencing. Integrated YOLO object detection to identify and classify objects within high-change frames, generating annotated timeline graphs and actionable CSV data to eliminate hours of manual video review.
- *Secure Remote Local LLM Deployment.* Deployed a private, self-hosted Qwen2.5-1.5B LLM on my local NVIDIA GPU for secure, zero-cost inference. Architected a zero-trust network tunnel utilizing zrok2 and developed a custom full-stack authenticated web interface, bridging ML infrastructure with networking to enable seamless, private global access to the local model via mobile devices. [Demo](#)

RESEARCH EXPERIENCE

University of Southern California, Graduate Research Assistant **08/2021 – present**

- *Novel generative model for brain evolution.* First-author project developing a novel 3D hybrid ControlNet + diffusion model to synthesize high-resolution, participant-specific future brain MRIs from baseline scans. This generalizable generative framework enables individualized prognosis of aging-related brain diseases and demonstrates scalable diffusion-based approaches for longitudinal medical imaging [4].
- *High-resolution biological age prediction.* Independently architected and implemented a novel PyTorch DNN that achieves an 8× spatial resolution improvement and 38% reduction in prediction error over prior state-of-the-art brain age models. Designed for memory-efficient inference on large-scale MRI datasets. [1] [\[GitHub\]](#) [\[Demo\]](#)

- *Open-source global brain age estimation.* Designed and released a reproducible TensorFlow-based DNN for brain age prediction from structural MRI, with emphasis on generalizability across datasets and accessible, well-documented implementation. [\[GitHub\]](#)

University of Southern California, Predoctoral Research Assistant

08/2019 – 07/2021

- *Scalable brain segmentation pipeline.* Prototyped automated head CT segmentation across 160+ anatomical regions, enabling large-scale cross-sectional analysis of brain atrophy patterns in a low-risk dementia population. Designed for robustness and generalizability across heterogeneous clinical data [2].
- *Research infrastructure & large-scale compute.* Independently architected and administered HPE Linux servers and AWS cloud environments for neuroimaging research; optimized NVIDIA A100 GPU clusters for distributed deep learning training, ensuring cost-efficient utilization of compute resources for long-running research projects.

Vellore Institute of Technology, Undergraduate Research Assistant

01/2016 – 05/2017

- *Multi-modal sensor fusion for activity prediction.* Built an integrated hardware-software system combining Arduino sensors and Android application to collect and fuse physiological and motion data (temperature, heart rate, accelerometer), enabling real-time prediction of optimal yoga postures through machine learning.

SELECTED PUBLICATIONS

- [1] Nikhil N Chaudhari, Nahian F Chowdhury, et al. “Deep learning maps local brain aging in relation to cognition across human adulthood”. In: *Proceedings of the National Academy of Sciences* (2026). DOI: 10.1073/pnas.2532233123.
- [2] Nikhil N Chaudhari, Phoebe E Imms, et al. “Increases in regional brain volume across two native South American male populations”. In: *GeroScience* (2024), pp. 1–21. DOI: 10.1007/s11357-024-01168-2.
- [3] Kevin H Guo, Nikhil N Chaudhari, et al. “Anatomic Interpretability in Neuroimage Deep Learning: Saliency Approaches for Typical Aging and Traumatic Brain Injury”. In: *Neuroinformatics* (2024), pp. 1–16. DOI: 10.1007/s12021-024-09694-2.
- [4] Gengshuo Liu*, Nikhil N Chaudhari*, et al. “Controllable Generative Model for Brain Evolution”. In: *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. *Equal contribution. 2025, pp. 1–5. DOI: 10.1109/ICASSP49660.2025.10888742.

SELECTED AWARDS and PATENTS

Resource Award *National Artificial Intelligence Research Resource* (Award ID: NAIRR240132)

US Patent *Regional segmentation and parcellation of the human cerebral cortex from computed tomography.* (Application No. 18/794,766)

LEADERSHIP

Mentored undergraduate and graduate students through programs including IUSSTF-Viterbi, CURVE, and Viterbi Graduate mentorship, focusing on research training and transition support. Served USC community by organizing cultural events including Diwali as Program Manager for the Hindu Student Organization.